
Reading Positions Off a Genotype–Phenotype Model: A Negative Result on Attribution Reliability in a Multiplexed Splicing Assay

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Reading the top attributed positions off a trained sequence–activity model is stan-
2 dard practice for guiding RNA therapeutic design [14], and that community has
3 argued such claims are validated anecdotally [22]. We test the practice on a real
4 multiplexed splicing assay and report that it is not reproducible. Models whose
5 held-out accuracy is statistically indistinguishable assign different top-3 positions
6 for 43% of held-out sequences at the best-fitting cell, rising to 94% at the worst;
7 no cell escapes it. Tightening the accuracy match until 17 of every 18 surviving
8 pairs at that cell are discarded, 34 of 35 grid-wide, leaves it unchanged. We find
9 three compounding causes. The fitting objective does not identify the attribution,
10 so the choice among equally-accurate models is made by the initialization seed.
11 Attribution mass is concentrated, a median 98% of a sequence’s total in 3 of 9 po-
12 sitions at the best cell over ten seeds, and we show in closed form that under such
13 concentration a Spearman correlation of 0.71 is what *perfect* agreement already
14 produces, so the rank correlations normally used to check agreement cannot distin-
15 guish success from failure. And the criterion the field selects on does not constrain
16 the function: tied models differ pointwise on the scale of the assay’s own Poisson
17 noise, so the 1000 held-out points the accuracy test already consumes determine
18 which of them generated the data. The test compresses them to one scalar first,
19 so the information is discarded rather than missing. We also report two ways our
20 own validation nearly hid the result: a control that read as passing because the
21 refit began at its own target, and a negative from a search that examined zero can-
22 didates. This is a reproducibility failure, not evidence that the attributed positions
23 are biologically wrong. We report the conditions under which it is worst, having
24 found none under which the practice is safe, and a four-item checklist.

25 1 Problem

26 **Biological task.** Given a short RNA sequence and a measured activity, identify *which positions*
27 *determine that activity*. In splice-site and siRNA design this is the object that guides the next round
28 of synthesis: a designer who reads positions 2, 5 and 6 as critical will vary those and hold the rest
29 fixed.

30 **Data modality and setting.** A multiplexed assay of variant effect: each sequence is assayed in
31 a pooled library and read out as a ratio of two sequencing count pools. We use the BRCA2 exon
32 17–19 5’ splice-site assay of Wong et al. [20], which quantifies $\sim 30,483$ nine-nucleotide sites with
33 $\log_{10}$ percent-spliced-in readouts and retains per-sequence read counts. We subsample $n = 4000$
34 per seed, split 3000 fit / 1000 held out by a fixed seeded permutation recorded with each run.

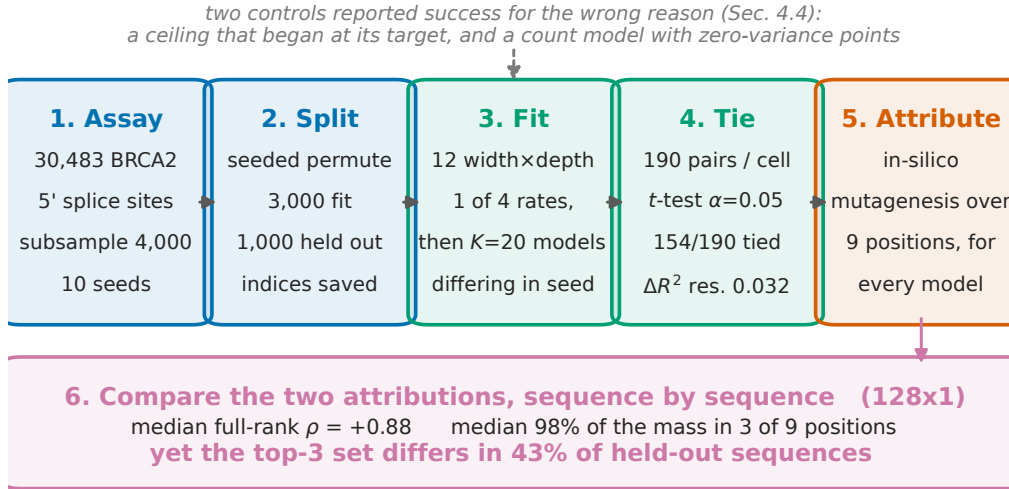


Figure 1: **The pipeline, and where the disagreement appears.** Every count is read from the committed run behind this paper. Models are trained per width $\times$ depth cell differing *only* in initialization seed, then filtered to the pairs a paired t -test on held-out squared errors cannot separate at $\alpha = 0.05$; the surviving pairs are compared by in-silico mutagenesis, sequence by sequence. At the best-fitting cell the survivors agree well on the full-position ranking and concentrate almost all their mass in three positions, yet they disagree about *which* three for nearly half of held-out sequences. The two controls named above the pipeline both returned a clean-looking result without testing what they were meant to test (Section 4, Table 8).

35 **Assumption under test.** That the top- k attributed positions recovered from a well-fitting model
 36 are a property of the sequence–activity relationship rather than of the particular model that happened
 37 to be trained.

38 **Target metric.** Agreement of the top- k attributed position *set* between models that *predictive accu-*
 39 *racy* cannot distinguish, measured per-sequence rather than averaged over the library. That qualifier
 40 is the subject of this paper: accuracy-matching is a statement about one aggregate scalar, not about
 41 the functions being close. We use $k = 3$ throughout, justified in Section 4 and by Proposition 1.

42 **Desired outcome.** For the practice to be sound, accuracy-matched models should agree on the
 43 top- k set for nearly all sequences: a designer should not get a different answer from a different seed.
 44 If they disagree, either the assay cannot tell the models apart, or it can and the selection procedure
 45 discards information the data already holds. We find the second.

46 2 Proposed approach

47 The approach under investigation is standard practice [2, 19], applied in genomics with in-silico
 48 mutagenesis since [21] and still standard [3, 11, 17], not a method we invented: fit a genotype–
 49 phenotype model, compute per-position attributions by in-silico mutagenesis, and read the top-
 50 ranked positions as the determinants.

51 **Core mechanism.** A model f_θ is fit by minimizing predictive loss on the assay. Attribution scores
 52 each position by the mean absolute change in prediction under substitution, over the real four-letter
 53 alphabet:

$$A_j(x) = \frac{1}{3} \sum_{b \neq x_j} |f_\theta(x^{j \rightarrow b}) - f_\theta(x)|. \quad (1)$$

54 Positions are ranked by A_j and the top k reported.

matching rule	pairs/seed	top-1 differs	top-2 differs	top-3 differs
accuracy-tied (t -test)	154	18%	29%	43% [41, 45]
$ \Delta R^2 < 0.01$	83	18%	29%	42% [40, 45]
$ \Delta R^2 < 0.005$	43	18%	30%	43% [41, 46]
$ \Delta R^2 < 0.001$	9	19%	29%	43% [41, 46]

Table 1: **The disagreement does not come from pairs that merely slipped past a t -test: tightening the accuracy match by a factor of 18 leaves it statistically unchanged.** Best-fitting cell (128×1); each row re-reads the same per-pair records under a stricter cap on the held-out R^2 gap. Entries are the mean over the 10 data-split seeds of that seed’s median over qualifying pairs, bootstrapped over seeds rather than pairs, since pairs within a seed share a split. The top-1 column shows the disagreement is not confined to a negligible third position. All twelve cells: Table 6.

Preconditions. (i) The model fits the assay well enough that its sensitivities are meaningful. (ii) The attribution is faithful to the model, in the sense that it reflects the model’s own dependence on each position [10]. (iii) Implicitly, that the model is *the* model: that the fitting procedure selects a function determined by the data.

Hypothesis under which it works. If the data determines f up to predictive equivalence, and attribution is faithful to f , then the top- k set is determined by the data. Precondition (iii) is the one we test, and it is the one that is almost never stated.

Models and protocol. Feedforward genotype–phenotype maps over a grid of widths $\{16, 32, 64, 128\}$ and depths $\{1, 2, 3\}$, with the per-cell learning rate chosen by pilot on the fit split. Within a cell we train $K = 20$ models differing *only* in initialization seed: same architecture, same split, same budget, same learning rate. The occurrence grid, the twin comparison and the cold-start control each use ten seeds; the per-instance dump behind Figure 4 uses five. Bootstrap intervals are over seeds; enumerations are exact and reported as exact. Claims and thresholds were pre-registered and committed before any run existed, and their single-commit history predates the earliest run directory (Appendix C). Total compute is on the order of GPU-days on a single mid-range machine.

3 Observed outcome

Models that predictive accuracy cannot separate assign different top-3 positions for 43% to 94% of held-out sequences, yet for the median such pair at most three held-out measurements determine which model generated the data. This mirrors the disagreement problem documented for general-purpose explainers [12], and joins results showing genomic sequence models can score well while generalizing poorly [16]. **The practice fails by discarding recoverable information.** Selection interrogates one aggregate scalar, and two models can sit inside its resolution while predicting point-wise differences on the scale of the assay’s noise. Whichever model the seed produced is then read for attributions as though the data had chosen it.

Accuracy matching. We filter to pairs whose held-out performance a paired t -test on per-point squared errors cannot separate at $\alpha = 0.05$. This is a failure to reject, not a demonstration of equivalence, so we report the filter’s resolution alongside it: the minimum detectable held-out R^2 difference at 80% power given $n_{\text{held-out}} = 1000$. The grid holds 2280 pairs per seed, of which 1610 are not separable at this resolution; pooled over the ten seeds that is 22,800 and 16,102. The per-cell means in Table 5 are rounded and sum to 1611.

Tightening the match does not change the result. Re-reading the same per-pair records under stricter caps on the held-out R^2 gap itself, moving from the t -test to $|\Delta R^2| < 0.001$ discards 17 of every 18 surviving pairs and leaves top-3 divergence at 43%, inside the unrestricted interval (Table 1). The disagreement is not carried by pairs that differ in predictive quality and merely escaped detection.

It is not an artifact of weak models. Held-out R^2 does correlate with agreement across cells (Spearman $+0.650$ against the minimum ρ), so we state the result at the *best*-fitting cell rather than

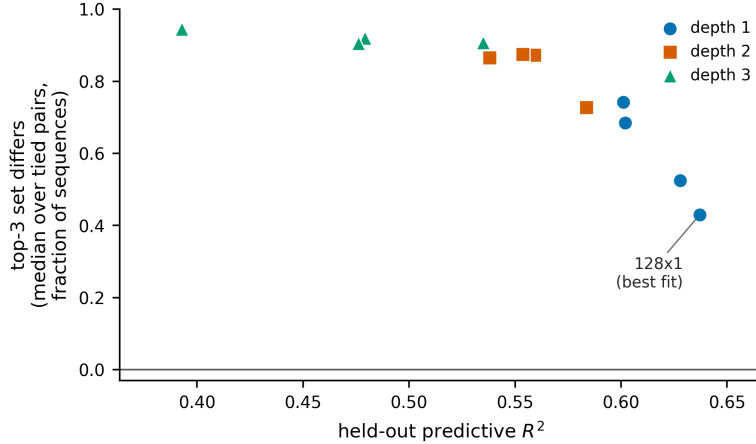


Figure 2: **The failure is not an artifact of poorly-fitting models.** Each point is one architecture cell: held-out predictive R^2 against top-3 attributed-set divergence between accuracy-matched models. **The plotted quantity is the median, over accuracy-tied pairs, of that pair’s fraction of held-out sequences whose top-3 sets differ**, a median over pairs of a per-pair fraction, not a single fraction pooled over sequences; the same definition is used in Tables 3 and 5. The best-fitting cell still diverges on 43% of sequences, so the failure cannot be dismissed as a symptom of underfitting.

cell	held-out R^2	tied pairs	resolution (ΔR^2)	ρ p10 / median	frac $\rho < 0.5$
16×1	0.6011	162/190	0.0699	+0.217 / +0.650	30%
16×2	0.5591	158/190	0.1059	-0.167 / +0.417	58%
16×3	0.5349	134/190	0.1161	-0.300 / +0.283	69%
128×1	0.6372	154/190	0.0318	+0.700 / +0.883	2%
128×2	0.5836	144/190	0.0461	+0.400 / +0.750	16%
128×3	0.4761	95/190	0.0937	-0.267 / +0.317	67%

Table 2: **Models the held-out data cannot tell apart still rank a sequence’s positions differently.** Accuracy-tied pairs and their attribution agreement. A pair is *tied* when a paired two-sided t -test on per-point held-out squared errors does not reject at $\alpha = 0.05$; *resolution* is the smallest held-out R^2 gap that test could have detected at 80% power, so a tie is a failure to reject at a stated resolution rather than a claim of equality. Per-instance ρ is a Spearman correlation between two models’ 9-position attribution vectors *for a single held-out sequence*, pooled over sequences and over tied pairs. **Rows are the two extreme widths at each depth**, which bracket both axes the argument turns on; all twelve cells, with bootstrap intervals, are in Table 5.

the worst. At width 128, depth 1, whose held-out $R^2 = 0.637$ is the highest in the grid, 154 of 190 pairs per seed are accuracy-tied, and among them 37% rank positions at ρ below 0.9, 15% below 0.8, and 6% below 0.7, reaching +0.350 (Figure 2). **Those are fractions of tied pairs**, one Spearman correlation each between the models’ library-averaged 9-position profiles. Table 5’s ρ column counts *per-sequence* correlations over (sequence, pair) observations instead, 10th percentile +0.700, as does Figure 4b, where 44,504 of 329,600 fall below 0.708. The denominators are not comparable.

The top- k set, which is what gets used. Prose and Tables 3 and 5 report the *pooled* estimator, one median over every tied pair with seeds pooled; Table 6 alone reports the *per-seed* one, and the two differ by up to a point. The top-3 set differs, pooled, for 73% and 90% of held-out sequences at width 128, depths 2 and 3; restricted to sequences where the concentration diagnostic (Section 4) says the comparison is meaningful, 71% and 80%. That conditioned figure carries little weight at depth 3, where only 11% of sequences at width 128 and 0.3% at width 64 have a well-defined top-3 under both models.

cell	top-3 mass	effective positions	top-1 differs	top-3 exact match	mean Jaccard
16×1	0.926	3.03	45%	26%	0.57
16×2	0.819	4.44	60%	13%	0.43
16×3	0.768	5.11	68%	9%	0.37
128×1	0.976	2.20	18%	57%	0.78
128×2	0.918	3.16	36%	27%	0.60
128×3	0.728	5.67	68%	10%	0.38

Table 3: **The disagreement is about which positions carry the mass, not about the ordering of a negligible tail.** Top-3 set agreement and attribution concentration. *top-3 mass* is the median fraction of a sequence’s total $|\Delta|$ in its three strongest positions and *effective positions* is exp of the entropy of the normalized magnitudes, where 9 means all positions contribute equally. Jaccard on 3-element sets takes only $\{0, 0.2, 0.5, 1\}$, so its median is near-useless as a summary and the *mean* is reported, alongside the fraction of sequences whose top-3 sets match exactly. **top-1 differs** is the fraction of sequences on which the two models’ single *strongest* position is a different position, a stronger form of disagreement than a reordering beneath a shared leader, and not implied by the top-3 columns, which are over unordered sets. Rows are the same bracketing subset as Table 2; all twelve cells are in Table 5.

Dominant position, or the order beneath it? A shuffle of ranks 2 and 3 is a weaker finding than a change of leader, and the top- k Jaccard cannot separate them: over unordered sets, two models can select the same three positions and still disagree about which dominates. We therefore report the strongest position separately (Tables 3 and 1). **The two models pick a different strongest position for 18% of sequences at width 128 depth 1, rising to 68% at depth 3;** the top-2 set differs for 29%, so the disagreement is not confined to a negligible third position. Averaged over all 329,600 sequence-pair observations, a third estimator, set disagreement at that cell is 44.6%, of which 35.8 points agree on the leader and differ only at ranks 2–3. **So at the best-fitting cell most of the top-3 disagreement is about ranks 2 and 3, and we say so rather than claim the stronger result;** at depth 3 the balance inverts, 63.3 of the 90.5 points also involving a different leader. These are accuracy-tied pairs of *independently trained* models, the population described in Section 2. A fit against its own reparameterized twin gives, for the same four quantities at width 128, a top-3 set divergence of 46% at depth 2 and 47% at depth 3 against the 73% and 90% above, and a conditioned divergence of 34% and 21% against 71% and 80%. The twin diverges less in eleven of twelve cells but *more* at the best-fitting cell, 46.1% against 43.0% (coincidentally rounding to the same 46% as depth 2). The two populations must not be quoted interchangeably; the twin is reported separately in Appendix G.

How cheaply the data adjudicates. Proposition 3 gives the measurements needed to determine which of two accuracy-tied models generated the data, under a noise scale derived from the assay’s own read counts (Proposition 2). Every requirement is reported as a whole number of measurements with a floor of one (Remark 3): the median is 1 to 3 in every cell, four orders of magnitude below the 30,483-sequence library (Figure 3). The quantity the bound divides by, $(\bar{d}_i/\sigma_i)^2$, has a per-cell median over tied pairs of 4.0 to 26.4, and is not tail-dominated: its mean exceeds its median by only 1.05 to 1.21 $\times$. Inverting those medians gives 1 to 2; the 3 is width 128 depth 1, where per-seed medians are averaged before ceiling and averaging does not commute with the reciprocal. **No new data is needed to act on this.** The evidence is already in hand: the same 1000 held-out points the accuracy test consumes. That test compresses them into one aggregate scalar before comparing, so the models are separated at the resolution of the scalar rather than of the points. The information is held and discarded, not missing. The complementary case, a fit against its own *reparameterized twin* needing up to 8.8×10^7 measurements yet diverging on top-3 anyway, is a different population, reported in Appendix G.

4 Reason for failure

4.1 Cause 1: the fitting objective does not identify the attribution

Predictive loss constrains f on the observed points, not which of the many equally-accurate functions is selected; that is settled by the initialization seed (Remark 1). This is the Rashomon ef-

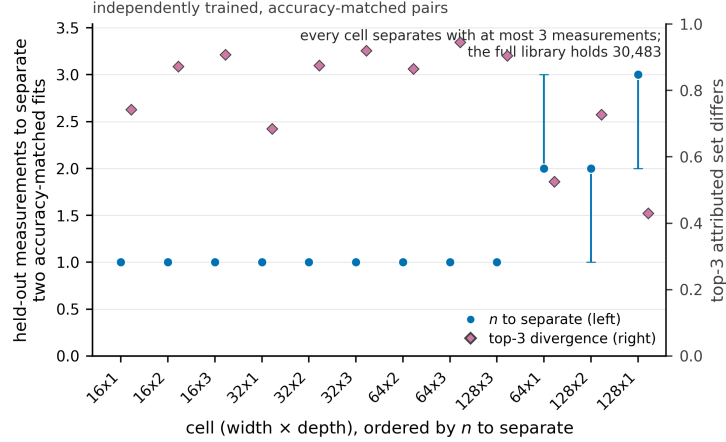


Figure 3: **Accuracy-tied models are cheap to tell apart as functions, yet disagree about which positions matter.** Held-out measurements required to determine which of two *independently trained* accuracy-matched models generated the data (Proposition 3), under the assay’s own Poisson count-noise model, with the top-3 set divergence on the right axis. Every cell needs at most three measurements, against a library of 30,483, so the reference line is off scale and is stated in the panel rather than drawn. The companion figure for a fit against its own reparameterized twin, where the requirement spans nearly eight orders of magnitude, is Figure 5.

fect [5, 7, 9, 13, 18] on a real assay, and the underspecification documented across ML pipelines [6]. What is new is not that multiplicity exists but its magnitude *at the level of the top- k set*, and that it is *not* forced by the data, which is the point this paper turns on.

The selection criterion is too coarse, which is a different problem from too little data. The accuracy filter cannot resolve a held-out R^2 gap below 0.032 even at the best-fitting cell, yet two models inside that gap are separable as functions by at most three measurements (Section 3). The held-out set already holds that evidence, the same 1000 points the t -test consumes, but the test reduces them to one aggregate scalar before comparing, so the information is discarded at the selection step rather than absent from the assay. Had the models been indistinguishable the remedy would be a larger assay; because they are not, it is to compare functions rather than aggregate error (Section 4.6), or to prefer a model interpretable by construction [15].

Precondition (iii) of Section 2 is therefore false, and it is the one nobody states. Faithfulness (precondition ii) does not help: a perfectly faithful attribution of an arbitrarily-selected model is an arbitrary answer.

4.2 Cause 2: attribution is concentrated, so the usual check cannot detect the failure

Attribution is not spread evenly across positions, and how unevenly depends strongly on the cell. The median fraction of a sequence’s $|\Delta|$ in its three strongest positions runs, over ten seeds, from 54% to 98% (width 128 depth 1), grid median 86%; effective positions run 7.80 to 2.20 of 9, grid median 3.92 (Table 3, Proposition 4). Figure 4a shows the most concentrated cell, width 128 depth 1: a median 98% in three positions, 0.71 at the single strongest, over the five seeds behind that figure and over all ten alike. Quote a per-cell value or a grid median deliberately, since they differ by more than ten points, and match the population: the twin comparison gives a grid median of 82% and 4.38.

This makes the standard check uninformative, and we can say by how much. Attribution is known to be fragile to small perturbations [8], and there are limits on what any attribution method can certify [4]; the point here is narrower: the statistic used to *check* agreement cannot see the failure. Proposition 1 shows that if two models agree *exactly* on the top 3 of 9 positions and order the remaining 6 at random, the expected Spearman correlation is 0.708. A reported ρ of 0.7 is therefore consistent with perfect agreement and with substantial disagreement alike.

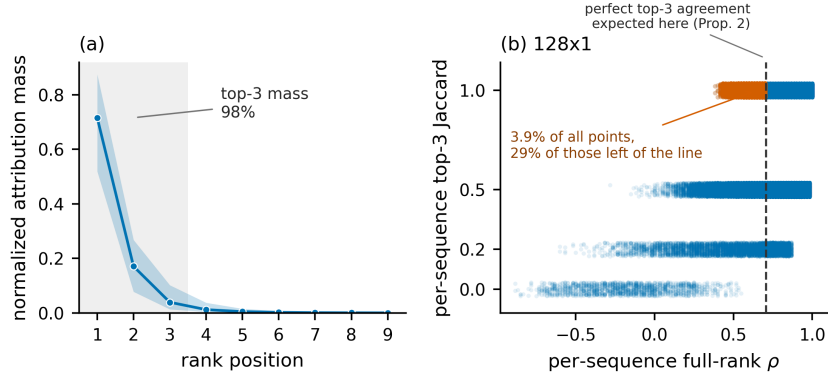


Figure 4: **A low rank correlation has two incompatible readings, and the top- k set separates them.** Accuracy-tied independently trained pairs at width 128, depth 1; five seeds, 329,600 (sequence, pair) points from 824 tied pairs. Those seeds pass the same $\alpha = 0.05$ filter as everywhere else and average 165 tied pairs against the ten-seed mean of 154. (a) Normalized attribution mass by rank position, median over 40,000 held-out sequence profiles with interquartile band: a median 98% of the mass sits in the top three positions and the remainder collapses to near zero. (b) Per-sequence full-rank ρ against per-sequence top-3 set Jaccard. The dashed line is $\rho = 0.708$, the value Proposition 1 gives for two models that agree *exactly* on the top 3 and order the remaining 6 at random. Points left of it with Jaccard = 1 (highlighted) are cases where the two models pick the same three positions while the rank correlation reads as disagreement. **There are 12,819 such (sequence, pair) observations: 3.9% of all 329,600 plotted, and 29% of those left of the line.** Each observation is one held-out sequence under one accuracy-tied pair, so the region is a substantial population rather than a handful of outliers. The statistic cannot distinguish the two readings; the top- k set can.

Diagnostic. Report the top- k position *set* overlap and the concentration alongside it. The conditioned form of that diagnostic rests on a usable population in eleven of twelve cells (at width 64 depth 3 only 0.3% of sequences have a well-defined top-3 under both models), while the unconditioned disagreement is present in all twelve. Exact top-3 agreement runs 26–57% at depth 1, 12–27% at depth 2 and 6–10% at depth 3, and the best cell still assigns a different top-3 set for 43% of sequences. We expected a low-capacity exemption and found none (Section 4.5).

4.3 Cause 3: the noise model, and a silent failure in ours

Asking how much data would separate two models requires a noise scale, and this assay supplies one. The phenotype is affine in $\log_{10}$ of a count ratio: regressing y on $\log_{10}(\text{ex}/\text{tot})$ gives slope 0.933 with $r = 0.990$. Those are two independent count pools, not a proportion, since ex exceeds tot in 4.9% of rows by up to $20.9\times$, so a binomial model is wrong here (Proposition 2).

Our first noise model failed silently, in the direction that would have overstated the result. Under a binomial form a near-saturated sequence receives $\sigma \rightarrow 0$; since Proposition 3 weights points by $1/\sigma^2$, a handful of such points dominated and the required sample size collapsed to zero: the analysis reported the models trivially separable. That model is also inapplicable on its own terms: $N_{\text{ex}} \geq N_{\text{tot}}$ on 4.9% of rows, where its variance is zero or negative. Under the Poisson log-ratio the weights are unremarkable (mean $1/\sigma^2$ is 3.0 times its median). We quote no magnitude for the binomial inflation: that model was replaced before the separation experiment was committed, so it cannot be regenerated from released code (Appendix F).

4.4 Two ways our validation nearly hid the failure

Both are errors we committed and caught, reported because the controls that catch them are cheap and appear not to be standard, in the spirit of the randomization checks that exposed saliency methods insensitive to their own model [1]. Table 8 states the ceiling control and the noise model of

193 Section 4.3 as what was reported, what was actually tested, and what the corrected check reports.
 194 The empty-search failure changed a verdict rather than a measurement, so it has no row.

195 **A ceiling that was arithmetic, and a negative from an empty search.** To check that a refit
 196 can recover a target, one refits to a target the class provably contains. Ours warm-started from
 197 the reference weights, so for the null target, the reference’s own latent, the initial loss was already
 198 6.7×10^{-19} , below the early-stopping threshold; the loop never ran and the control returned 1.000000
 199 because it *began at the answer*. It verified that the optimizer does not wander away from a perfect
 200 solution, never that the procedure can *find* one it does not start at. Cold-starting the same refit is a
 201 ceiling that requires search, and under it recovery of a target the architecture realizes *exactly* falls to
 202 held-out $R^2 \approx 0.45$ at depth 3 (Appendix H). Separately, an experiment reported “no witness found”
 203 after examining zero candidates, every cell having had an empty candidate set for structural reasons.
 204 Searched-and-found-nothing is a result; never-searched is not, and the second was reported as the
 205 first. We now emit a VOID verdict whenever a negative would be drawn from an empty candidate
 206 set, with a regression test pinning the distinction.

207 4.5 Boundary conditions

208 We found no regime in this grid where the practice is safe. What we can give is the shape of the
 209 degradation.

- 210 • **Depth is what tracks the failure; width is not, and no cell is safe.** Pooled top-3 di-
 211 vergence rises with depth: 43–74%, 73–88%, 90–94%, rank-correlated +0.92 with depth
 212 against −0.28 with width. Within depth 1 it improves with width, 74% to 43%, but never
 213 falls below 43%: we looked for a low-capacity exemption and found the opposite.
- 214 • **Fit quality predicts agreement but does not explain it.** Held-out R^2 correlates +0.650
 215 with agreement: real, far from explanatory, and the best-fitting cell still fails.
- 216 • **Pairs that are harder to tell apart agree more, not less.** Across cells the *unfloored* mea-
 217 surements needed to separate a tied pair are rank-correlated −0.94 with top-3 divergence,
 218 the floor having collapsed nine cells onto the same value: fits closer as functions pick more
 219 similar positions, the opposite of what the twin comparison shows (Appendix G). The fail-
 220 ure is not a by-product of the two fits being nearly identical.
- 221 • **The grid does not reach the noise ceiling.** The assay’s own count noise caps achievable
 222 held-out R^2 at $1 - \overline{\sigma_i^2} = 0.740$ [0.727, 0.749], against 0.637 at our best cell. That uses the
 223 mean of σ_i^2 ; squaring the median scale of Appendix F instead gives 0.903, higher because
 224 the noise is right-skewed. The grid sits below the ceiling, so this result shows the failure is
 225 not confined to the worst-fitting models rather than that it survives at the ceiling. Counting
 226 noise is a lower bound, so the true ceiling is lower and the gap narrower, but we have not
 227 closed it.
- 228 • **Scope.** One assay, one phenotype, sequence length 9, and each fit draws 4000 of the li-
 229 brary’s 30,483 sequences, so we cannot say whether the instability persists at full-library
 230 scale. The class tested is a one-hot MLP with a MAVE-NN-style monotone link, not the
 231 convolutional architectures the practice is usually run on, and we tested only in-silico mu-
 232 tagenesis. This is a reproducibility failure, the same data and procedure returning different
 233 top- k sets, not evidence that any attributed position is biologically wrong.

234 4.6 Actionable takeaways

- 235 1. **Report per-instance top- k sets, not a library-averaged rank correlation**, which by
 236 Proposition 1 cannot distinguish perfect from substantial disagreement, **and report the**
 237 **concentration of the attribution**, without which that correlation has no reference value.
- 238 2. **Derive the noise scale from the assay where counts are available** and check the count
 239 model; ours would have overstated the result.
- 240 3. **Train more than one seed and report top- k set agreement across them:** a constant-
 241 factor cost, and the cheapest check that catches this.
- 242 4. **Verify that a passing control required a search, and refuse to report a negative from**
 243 **an empty candidate set.**

- 245 [1] Adebayo, J., Gilmer, J., Muelly, M., Goodfellow, I., Hardt, M. & Kim, B. (2018) Sanity checks for saliency
246 maps. In *Advances in Neural Information Processing Systems 31*. arXiv:1810.03292.
- 247 [2] Alipanahi, B., Delong, A., Weirauch, M.T. & Frey, B.J. (2015) Predicting the sequence specificities of DNA-
248 and RNA-binding proteins by deep learning. *Nature Biotechnology* **33**(8):831–838.
- 249 [3] Avsec, Ž., Weilert, M., Shrikumar, A., Krueger, S., Alexandari, A., Dalal, K., Fropf, R., McAnany, C.,
250 Gagneur, J., Kundaje, A. & Zeitlinger, J. (2021) Base-resolution models of transcription-factor binding reveal
251 soft motif syntax. *Nature Genetics* **53**:354–366.
- 252 [4] Bilodeau, B., Jaques, N., Koh, P.W. & Kim, B. (2024) Impossibility theorems for feature attribution. *Pro-
253 ceedings of the National Academy of Sciences* **121**(2):e2304406120.
- 254 [5] Breiman, L. (2001) Statistical modeling: the two cultures. *Statistical Science* **16**(3):199–231.
- 255 [6] D’Amour, A., Heller, K., Moldovan, D., Adlam, B., Alipanahi, B., Beutel, A., et al. (2020) Underspecifica-
256 tion presents challenges for credibility in modern machine learning. arXiv:2011.03395.
- 257 [7] Fisher, A., Rudin, C. & Dominici, F. (2019) All models are wrong, but many are useful: learning a variable’s
258 importance by studying an entire class of prediction models simultaneously. *Journal of Machine Learning
259 Research* **20**(177):1–81.
- 260 [8] Ghorbani, A., Abid, A. & Zou, J. (2019) Interpretation of neural networks is fragile. In *Proceedings of the
261 AAAI Conference on Artificial Intelligence* **33**:3681–3688.
- 262 [9] Hsu, H. & Calmon, F.P. (2022) Rashomon capacity: a metric for predictive multiplicity in classification. In
263 *Advances in Neural Information Processing Systems 35*, pp. 28988–29000.
- 264 [10] Jacovi, A. & Goldberg, Y. (2020) Towards faithfully interpretable NLP systems: how should we define
265 and evaluate faithfulness? In *Proceedings of the 58th Annual Meeting of the Association for Computational
266 Linguistics*, pp. 4198–4205.
- 267 [11] Koo, P.K. & Eddy, S.R. (2019) Representation learning of genomic sequence motifs with convolutional
268 neural networks. *PLOS Computational Biology* **15**(12):e1007560.
- 269 [12] Krishna, S., Han, T., Gu, A., Wu, S., Jabbari, S. & Lakkaraju, H. (2022) The disagreement problem in
270 explainable machine learning: a practitioner’s perspective. arXiv:2202.01602.
- 271 [13] Marx, C., Calmon, F. & Ustun, B. (2020) Predictive multiplicity in classification. In *Proceedings of the
272 37th International Conference on Machine Learning*, PMLR **119**:6765–6774.
- 273 [14] Novakovsky, G., Dexter, N., Libbrecht, M.W., Wasserman, W.W. & Mostafavi, S. (2023) Obtaining genet-
274 ics insights from deep learning via explainable artificial intelligence. *Nature Reviews Genetics* **24**:125–137.
- 275 [15] Rudin, C. (2019) Stop explaining black box machine learning models for high stakes decisions and use
276 interpretable models instead. *Nature Machine Intelligence* **1**:206–215.
- 277 [16] Sasse, A., Ng, B., Spiro, A.E., Tasaki, S., Bennett, D.A., Gaiteri, C., De Jager, P.L., Chikina, M. &
278 Mostafavi, S. (2023) Benchmarking of deep neural networks for predicting personal gene expression from
279 DNA sequence highlights shortcomings. *Nature Genetics* **55**:2060–2064.
- 280 [17] Seitz, E.E., McCandlish, D.M., Kinney, J.B. & Koo, P.K. (2024) Interpreting cis-regulatory mechanisms
281 from genomic deep neural networks using surrogate models. *Nature Machine Intelligence* **6**:701–713.
- 282 [18] Semenova, L., Rudin, C. & Parr, R. (2022) On the existence of simpler machine learning models. In
283 *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency*, pp. 1827–1858.
- 284 [19] Tareen, A., Kooshkbaghi, M., Posfai, A., Ireland, W.T., McCandlish, D.M. & Kinney, J.B. (2022) MAVE-
285 NN: learning genotype–phenotype maps from multiplex assays of variant effect. *Genome Biology* **23**:98.
- 286 [20] Wong, M.S., Kinney, J.B. & Krainer, A.R. (2018) Quantitative activity profile and context dependence of
287 all human 5’ splice sites. *Molecular Cell* **71**(6):1012–1026.
- 288 [21] Zhou, J. & Troyanskaya, O.G. (2015) Predicting effects of noncoding variants with deep learning-based
289 sequence model. *Nature Methods* **12**:931–934.
- 290 [22] Zhou, S., Huang, M. & Li, K. (2026) Position: genomic model research must move beyond anecdotal eval-
291 uation of interpretability methods. In *Proceedings of the 43rd International Conference on Machine Learning*.
292 arXiv:2606.07607.

293 A Formal statements

294 The statements below are elementary. We give them explicitly because two of them (Propositions 1
295 and 2) determine how the main-text numbers should be read, and because the failure described in
296 Section 4.3 was a violation of Proposition 2 that produced a plausible-looking wrong answer rather
297 than an error.

298 **Remark 1** (Predictive risk does not determine the attribution). *Let $\mathcal{F}$ be a model class, R a popu-*
299 *lation risk, and $\mathcal{F}_\epsilon = \{f \in \mathcal{F} : R(f) \leq \min_{f'} R(f') + \epsilon\}$ the ϵ -Rashomon set. Let $A : \mathcal{F} \rightarrow \mathbb{R}^L$*
300 *be the attribution map of Eq. (1). Nothing in the definition of $\mathcal{F}_\epsilon$ constrains A to be constant on it.*
301 *Consequently, the attribution reported by a fitting procedure is a function of which element of $\mathcal{F}_\epsilon$*
302 *the optimizer returns, and that is determined by the initialization, not by the data. The empirical*
303 *content of this paper is the diameter of $A(\mathcal{F}_\epsilon)$ on a real assay, measured at the level of the top- k set.*

304 **Proposition 1** (Expected rank correlation under concentrated attribution). *Let two models rank L*
305 *positions. Suppose they agree exactly on the ordering of the top k positions, and that the remaining*
306 *$m = L - k$ positions carry negligible attribution mass, so that their relative orderings are inde-*
307 *pendent uniform random permutations. Then the expected Spearman rank correlation over all L*
308 *positions is*

$$\mathbb{E}[\rho_L] = 1 - \frac{m(m^2 - 1)}{L(L^2 - 1)}. \quad (2)$$

309 For $L = 9$ and $k = 3$ this is $1 - 2^{10}/720 = 0.708$.

310 *Proof.* Spearman's ρ over L items is $\rho = 1 - 6 \sum_i d_i^2 / (L(L^2 - 1))$, where d_i is the rank difference
311 at item i . The top k contribute $d_i = 0$ by assumption. For two independent uniform permutations of
312 m items, $\mathbb{E}[\rho_m] = 0$, and since $\rho_m = 1 - 6 \sum d_i^2 / (m(m^2 - 1))$ this gives $\mathbb{E}[\sum d_i^2] = m(m^2 - 1)/6$
313 over the tail block. Substituting yields the result. $\square$

top- k agreed	$k = 2$	$k = 3$	$k = 4$	$k = 5$
$\mathbb{E}[\rho_9]$	0.533	0.708	0.833	0.917

Table 4: Expected Spearman correlation over $L = 9$ positions when two models agree *exactly* on the top k and order the rest at random (Proposition 1). A reported ρ at or below these values carries no evidence of disagreement about which positions matter.

314 **Proposition 2** (Poisson log-ratio noise scale). *Let the phenotype be $y = a \log_{10}(N_{\text{ex}}/N_{\text{tot}}) + b$ with*
315 *N_{ex} and N_{tot} independent Poisson counts with means $\lambda_{\text{ex}}, \lambda_{\text{tot}}$. Then to first order*

$$\text{sd}(y) = \frac{|a|}{\ln 10} \sqrt{\frac{1}{\lambda_{\text{ex}}} + \frac{1}{\lambda_{\text{tot}}}} \approx \frac{|a|}{\ln 10} \sqrt{\frac{1}{N_{\text{ex}}} + \frac{1}{N_{\text{tot}}}}. \quad (3)$$

316 *Proof.* For $N \sim \text{Poisson}(\lambda)$, the delta method with $g(N) = \ln N$ gives $\text{Var}(\ln N) \approx$
317 $\lambda^{-2} \text{Var}(N) = 1/\lambda$. Independence of the two pools gives $\text{Var}(\ln N_{\text{ex}} - \ln N_{\text{tot}}) \approx 1/\lambda_{\text{ex}} + 1/\lambda_{\text{tot}}$.
318 Converting to base 10 and scaling by a yields the result; substituting counts for means is the usual
319 plug-in estimator. $\square$

320 **Remark 2** (Why the binomial form fails here, and fails silently). *Modelling the readout as a propor-*
321 *tion p from N trials gives $\text{sd}(y) \propto \sqrt{(1-p)/(Np)}$, which tends to 0 as $p \rightarrow 1$. Since the statistic*
322 *of Proposition 3 weights points by $1/\sigma_i^2$, a small number of near-saturated sequences then carry*
323 *unbounded weight. Under Proposition 2 the ratio $\text{mean}(1/\sigma^2)/\text{median}(1/\sigma^2)$ is 3.0, a heavy but*
324 *unremarkable tail. We give no corresponding figure for the binomial form: it was replaced before*
325 *the separation experiment was first committed and cannot be regenerated from released code, so*
326 *only the structural defect is quoted. That defect is decisive on its own: $N_{\text{ex}} > N_{\text{tot}}$ in 4.9% of rows,*
327 *by up to $20.9\times$, so the two counts are not numerator and denominator of a proportion.*

328 **Proposition 3** (Separation sample size). *Let f_1, f_2 be two candidate functions with pointwise differ-*
329 *ence $d_i = f_1(x_i) - f_2(x_i)$ at held-out points x_i , and let observations carry independent Gaussian*

noise with known standard deviations σ_i . Testing $H_0 : y \sim f_1$ against $H_1 : y \sim f_2$ at two-sided level α with power $1 - \beta$ requires

$$n \geq \frac{(z_{1-\alpha/2} + z_{1-\beta})^2}{(\overline{d_i/\sigma_i})^2}, \quad \overline{(d_i/\sigma_i)^2} = \frac{1}{n} \sum_{i=1}^n \frac{d_i^2}{\sigma_i^2}. \quad (4)$$

Proof. The log-likelihood ratio between the two simple hypotheses is Gaussian with mean $\pm \frac{1}{2} \sum_i d_i^2/\sigma_i^2$ under H_0 and H_1 respectively, and variance $\sum_i d_i^2/\sigma_i^2$. The standardized separation is therefore $(\sum_i d_i^2/\sigma_i^2)^{1/2}$, and the standard two-sided power requirement $(\sum_i d_i^2/\sigma_i^2)^{1/2} \geq z_{1-\alpha/2} + z_{1-\beta}$ rearranges to the stated bound. $\square$

Remark 3 (Whole measurements, with a floor of one). *The bound divides by $(\overline{d_i/\sigma_i})^2$, an average over points, so it can return a value below 1. Such a value does not name a number of measurements: it says the average standardized difference already exceeds the threshold at a single point. Zero is also unplotable on a log axis. Every requirement reported in this paper is therefore $\lceil n \rceil$ with a floor of 1.*

Remark 4 (The reported requirement is a lower bound). *Proposition 2 accounts for sequencing count noise only. Library preparation, transfection and biological variation add further variance, which increases σ_i and hence increases the required n . Every separation requirement in the main text is therefore a lower bound on the measurements needed.*

Proposition 4 (Effective number of contributing positions). *For an attribution vector $A \in \mathbb{R}_{\geq 0}^L$ with $\sum_j A_j > 0$, let $q_j = A_j / \sum_{j'} A_{j'}$. The effective number of contributing positions is $N_{\text{eff}}(A) = \exp(-\sum_j q_j \ln q_j)$, satisfying $1 \leq N_{\text{eff}} \leq L$, with $N_{\text{eff}} = L$ iff all positions contribute equally and $N_{\text{eff}} \rightarrow 1$ as one position dominates.*

This is the exponentiated Shannon entropy of the normalized attribution and is reported per-sequence in Table 3; the median over held-out sequences is 3.92 of 9 for accuracy-tied independently trained pairs and 4.38 for the twin comparison, both medians *across cells*; per cell the value ranges from 2.20 to 7.80, so the reference value in Proposition 1 must be read against the cell in question rather than against the grid median.

B Full per-cell results

Table 5 combines Tables 2 and 3 for every cell in the width $\times$ depth grid, with bootstrap intervals on the columns that admit them. It is generated by `experiments/make_tables.py` directly from the committed run directories, as are all tables in this paper.

Table 6 repeats the accuracy-match tightening of Table 1 in every cell. The pair population falls 35-fold across the grid and the top- k disagreement does not move anywhere.

C Protocol and pre-registration

Table 7 gives the per-cell protocol, read from the `tuning_budget.json` written with each run and from the per-seed records. The split rule is the same everywhere: for seed k a generator seeded with `1,000,003 + k` draws a permutation of the 4000 subsampled sequences, the first 3000 form the fit split and the remaining 1000 are held out, and the indices are recorded in the run directory. All twelve cells use ten seeds.

Architecture and optimizer. The latent map is a fully-connected stack of `depth` $\times$ [Linear, tanh] followed by a linear projection to a scalar, in `float64`, over a one-hot encoding of the nine-position site. Its output is passed through a MAVE-NN-style monotone global-epistasis nonlinearity: a sum of 5 sigmoids with non-negative weights, hence strictly increasing, fitted jointly with the latent map. All parameters are initialized i.i.d. $\mathcal{N}(0, 0.3^2)$ from the per-model seed, which is the *only* thing that differs between the $K = 20$ models in a cell. Training minimizes mean-squared error on the fit split with Adam, full-batch, for the fixed epoch budget in Table 7, with no weight decay, no dropout and no early stopping; the learning rate is the one chosen by the per-cell pilot. In-silico mutagenesis

(a) accuracy-tied pairs and per-instance agreement					
cell	held-out R^2	tied pairs	resolution	ρ p10 / median	frac $\rho < 0.5$
16×1	0.6011 [0.5725, 0.6274]	162 [156, 169]	0.0699	+0.217 / +0.650	30%
16×2	0.5591 [0.5243, 0.5880]	158 [147, 167]	0.1059	-0.167 / +0.417	58%
16×3	0.5349 [0.4948, 0.5677]	134 [122, 147]	0.1161	-0.300 / +0.283	69%
32×1	0.6019 [0.5754, 0.6263]	167 [160, 172]	0.0651	+0.365 / +0.717	19%
32×2	0.5537 [0.5110, 0.5909]	143 [128, 158]	0.1032	-0.168 / +0.408	58%
32×3	0.4792 [0.4128, 0.5349]	102 [97, 108]	0.1256	-0.317 / +0.258	71%
64×1	0.6278 [0.6003, 0.6544]	160 [151, 169]	0.0402	+0.600 / +0.833	4%
64×2	0.5381 [0.4846, 0.5861]	100 [81, 120]	0.0882	-0.183 / +0.417	57%
64×3	0.3929 [0.3309, 0.4483]	92 [86, 97]	0.1267	-0.383 / +0.175	78%
128×1	0.6372 [0.6122, 0.6585]	154 [138, 166]	0.0318	+0.700 / +0.883	2%
128×2	0.5836 [0.5369, 0.6259]	144 [119, 168]	0.0461	+0.400 / +0.750	16%
128×3	0.4761 [0.3949, 0.5456]	95 [71, 117]	0.0937	-0.267 / +0.317	67%

(b) top- k set agreement and concentration					
cell	top-3 mass	effective positions	top-1 differs	top-3 exact	mean Jaccard
16×1	0.926	3.03	45%	26%	0.57
16×2	0.819	4.44	60%	13%	0.43
16×3	0.768	5.11	68%	9%	0.37
32×1	0.956	2.56	37%	32%	0.62
32×2	0.851	4.05	62%	12%	0.42
32×3	0.759	5.28	69%	8%	0.36
64×1	0.969	2.36	24%	48%	0.73
64×2	0.873	3.79	61%	14%	0.43
64×3	0.540	7.80	74%	6%	0.31
128×1	0.976	2.20	18%	57%	0.78
128×2	0.918	3.16	36%	27%	0.60
128×3	0.728	5.67	68%	10%	0.38

Table 5: **No cell is withheld: the pattern in Tables 2 and 3 holds across the whole grid.** All twelve cells. The upper panel carries the columns of Table 2, the lower those of Table 3. **Which columns carry intervals, and why the rest do not:** held-out R^2 and the tied-pair count are per-seed quantities and are given as mean with a 95% bootstrap interval over the ten seeds. The remaining columns are medians pooled over every tied pair across all seeds rather than per-seed values, so a seed-level bootstrap interval is not defined for them and none is invented here.

width	depth	accuracy-tied (t -test)				$ \Delta R^2 < 0.001$			
		pairs	top-1	top-2	top-3	pairs	top-1	top-2	top-3
16	1	162	45%	62%	74% [73, 76]	6	43%	60%	72% [68, 76]
16	2	158	60%	79%	87% [86, 89]	3	58%	77%	86% [83, 88]
16	3	134	67%	84%	90% [88, 92]	2	65%	82%	88% [84, 91]
32	1	167	36%	55%	68% [67, 70]	5	35%	52%	66% [63, 69]
32	2	143	59%	80%	87% [86, 89]	3	62%	81%	88% [86, 90]
32	3	102	68%	86%	92% [91, 93]	1	69%	86%	92% [91, 94]
64	1	160	24%	38%	52% [47, 57]	6	24%	37%	49% [44, 54]
64	2	100	57%	79%	86% [82, 89]	2	53%	76%	85% [80, 90]
64	3	92	73%	89%	94% [94, 95]	1	67%	84%	90% [82, 94]
128	1	154	18%	29%	43% [41, 45]	9	19%	29%	43% [41, 46]
128	2	144	38%	61%	74% [71, 77]	6	29%	55%	72% [68, 76]
128	3	95	68%	86%	91% [89, 92]	2	63%	85%	92% [89, 94]

Table 6: Tightening the accuracy match to $|\Delta R^2| < 0.001$ in every cell. The pair population falls 35-fold across the grid, from 1,611 tied pairs per seed to 46, and the top- k disagreement does not move, so the failure is not an artifact of the t -test’s resolution. Columns as in Table 1; intervals bootstrap over data-split seeds. The top- k columns here are recomputed *per seed* and then averaged, which is why they carry seed-level intervals where the pooled columns of Table 5 cannot, and why a cell may differ from that table by a point.

cell	learning rate chosen (seeds)	configs	epochs	grad. steps
16×1	0.01	4	1,200	1,200
16×2	0.01	4	1,200	1,200
16×3	0.01 (×9), 0.003 (×1)	4	1,200	1,200
32×1	0.01	4	1,200	1,200
32×2	0.01	4	1,200	1,200
32×3	0.01	4	1,200	1,200
64×1	0.01 (×4), 0.003 (×6)	4	1,200	1,200
64×2	0.01 (×9), 0.003 (×1)	4	1,200	1,200
64×3	0.01 (×9), 0.003 (×1)	4	1,200	1,200
128×1	0.003 (×9), 0.001 (×1)	4	1,200	1,200
128×2	0.003 (×4), 0.001 (×6)	4	1,200	1,200
128×3	0.003 (×4), 0.001 (×6)	4	1,200	1,200

Table 7: **The tuning budget is four learning rates per cell and nothing else, so the disagreement cannot be an artifact of unequal search.** Per-cell protocol, read from `tuning_budget.json` and the per-seed records. The search space is “learning rate over [0.01, 0.003, 0.001, 0.0003], selected once per cell and shared by all models in it”. The selection rule is identical in every cell, “no model selection: every trained model is kept and enters the pairwise analysis”, so it is stated here rather than repeated as a column. **The chosen rate is not constant across seeds:** it varies in 7 of 12 cells, and the multiplicities are given. Within any one seed a single rate is shared by all 20 models in the cell, so the models still differ only in initialization; across seeds the rate is re-selected.

check	what it reported	what it actually tested	corrected control
Ceiling control	Held-out $R^2 = 1.000000$ recovering a target the architecture realizes exactly, in every cell and every seed.	That Adam does not leave a perfect solution. Warm-started at the target, its initial loss was already below the stopping threshold and best-iterate selection kept the initial state, so the loop never ran.	Cold-started on the same target, the worst cell reaches only 0.4501 and 11 of 12 cells fall below the 0.999 threshold.
Noise model	A required sample size of zero: any two fits separable with no data at all.	Inverse-variance weighting under a count model undefined on this library: the pools are not a proportion, $ex \geq tot$ on 4.9% of rows, so binomial variance is zero or negative there and a few near-saturated sequences carried the statistic.	Under the Poisson log-ratio the weights are ordinary (mean $1/\sigma^2$ is $3.0\times$ its median) and the requirement is finite and population-specific: 1 to 3 measurements for accuracy-tied independently trained pairs, and 1 to 87,751,464 for a fit against its reparameterized twin. The two must not be quoted as one range.

Table 8: **Both controls we relied on were passing for reasons unrelated to what they were meant to test.** Neither failure was visible from the reported number alone: a control that cannot fail and a weighting scheme that silently concentrates on a few points both look like success.

374 is evaluated on 400 held-out sequences per pair, so the per-sequence fractions in this paper have that
375 denominator.

376 Claims, thresholds and go/no-go criteria were written and committed before any result existed:
377 PREREGISTRATION.md has a single commit, dated 2026-08-26, and the earliest committed run di-
378 rectory is dated 2026-08-29, so the ordering is checkable from the repository history rather than
379 asserted. Amendment is additionally blocked by a pre-write hook in the authoring environment
380 that matches the target path of a write; we name the mechanism precisely because it constrains our
381 workflow rather than the repository, and the durable evidence is the single-commit history. Every
382 reported number carries the git SHA and configuration hash of the run that produced it.

D Validation controls

The mechanism, the per-cell numbers and both arms of the replacement control are given once, in Appendix H; they are not repeated here.

The general form of the error is worth stating separately from its instance. A control that verifies a procedure can recover a target must ensure the procedure does not *begin* at that target. Warm-starting a refit from the reference weights and then asking it to recover the reference’s own output satisfies the letter of the control and none of its purpose.

E Reproducibility

Code, pre-registered analysis plans, per-seed raw values and run provenance will be released on acceptance; the repository is withheld here for anonymity. Each result directory records the git SHA, the configuration hash, the seed list and the tuning budget consumed, and every table and figure in this paper is generated from those directories by scripts included in the release rather than transcribed by hand.

F Noise model derivation

Throughout this appendix Z denotes the number of held-out measurements needed to determine which of two candidate fits generated the data, at $\alpha = 0.05$ with power 0.8 (Proposition 3), reported as $\lceil Z \rceil$ with a floor of 1. It requires an observation-noise scale. This library supplies one rather than needing an assumption, and the derivation is given here because the choice of count model changes Z by orders of magnitude.

The phenotype is affine in a log count ratio. Each sequence carries two count pools, *ex* and *tot*. Regressing the phenotype y on $\log_{10}(\text{ex}/\text{tot})$ gives slope 0.9331 [0.9314, 0.9345] across seeds, and $r = 0.990$ over the full 30,483-sequence library. The slope is fitted per seed on that seed’s own 4000-row subsample and *excludes* the 11.4% of rows with $\text{ex} = 0$, where the log ratio is undefined; the half-count continuity correction below is applied to the noise scale, not to this regression, so it does not put those rows back. The correlation is quoted over the full library on the same $\text{ex} > 0$ rows. The relationship is tight enough that the count ratio, not the phenotype, is the quantity whose noise must be modelled.

Why Poisson and not binomial. The two pools are *not* numerator and denominator of a proportion. Over the committed library, *ex* exceeds *tot* in 4.9% of rows, with ratios as large as 20.9, and $\text{ex} = 0$ in 11.4% of rows. A binomial model is therefore not merely inefficient but undefined on roughly one row in twenty: $p = \text{ex}/\text{tot} \geq 1$ there, so the binomial variance $p(1-p)/\text{tot}$ is zero or negative.

Delta method. Treating both pools as independent Poisson variates and applying the delta method to the log of their ratio,

$$\text{var}(\log_{10} R) = \frac{1}{\ln(10)^2} \left(\frac{1}{\text{ex}} + \frac{1}{\text{tot}} \right), \quad \text{sd}(y_i) = |a| \frac{\sqrt{1/\text{ex}_i + 1/\text{tot}_i}}{\ln 10}, \quad (5)$$

with a the fitted slope and a half-count continuity correction applied to both pools, which also covers the rows with $\text{ex} = 0$. In the standardized units the models see, the median of this quantity is 0.3114 [0.3051, 0.3194].

The diagnostic that surfaced the binomial failure. The separation test weights points by $1/\sigma_i^2$, so a point assigned near-zero noise dominates the statistic. Under a binomial model a near-saturated sequence receives exactly that. The symptom was a weight distribution with a mean far above its median and a required sample size that collapsed to zero for every cell, an implausible answer that is what prompted the check. Under the committed Poisson log-ratio model the mean of $1/\sigma^2$ is 3.0 times its median, a heavy but unremarkable tail, and Z becomes finite and cell-dependent. An unexamined count model would have reported that no data at all is needed to separate the fits.

We deliberately do not quote a magnitude for the binomial inflation. That model was replaced before the first commit of the separation experiment, so its exact variance form is not in the repository and the figure cannot be regenerated from committed code. What *is* reproducible from the committed library is the structural defect above, $p \geq 1$ on 4.9% of rows and hence zero or negative binomial variance, together with the Poisson ratio of 3.0. Both were re-derived from the committed 30,483-row library while preparing this appendix and agree with the recorded values.

A figure comparing the two weight distributions was considered and cut. It would need the binomial variance form, which is not in the repository, and per-observation $1/\sigma^2$ values, which no run directory stores, since only the median noise scale survives into `results.json`. Drawing it would have meant inventing the comparator and recomputing the rest outside any recorded run, so the comparison stays in prose where its provenance can be stated.

Z is a lower bound. This models sequencing noise only. Library preparation and biological variation add more, so the true noise is larger and the required sample size larger still. Every Z in this paper should be read as “at least this many”.

G Separability against a reparameterized twin

The main text measures *independently trained* models that predictive accuracy cannot separate, and shows that the assay itself separates them cheaply, in at most three held-out measurements. This section reports the complementary case: a single fit against a *reparameterized twin* of itself. The cost of adjudication varies over nearly eight orders of magnitude across the grid: it is under ten measurements at every depth-1 cell, and exceeds the full library only at width 128 depths 2 and 3. Those two cells are the ones where the data genuinely cannot adjudicate.

The two cases bracket the problem. In the main-text population the divergence is avoidable: for the median tied pair one to three of the held-out points already in hand settle which of the two generated the data, and the selection procedure never asks. Here it is not avoidable, since at width 128 depth 3 the twin needs 8.8×10^7 measurements, more than three orders of magnitude beyond the full library, and the top-3 sets diverge regardless. Attribution therefore fails in both regimes, for different reasons, and only the first has a cheap remedy.

How the twin is constructed, and what its R^2 measures. The reference fit’s latent $\hat{\phi}$ is passed through a fixed strictly monotone reparameterization ψ , and a second network of the same architecture is refit to reproduce $\psi(\hat{\phi})$ on the fit split. Its phenotype prediction is its own latent put back on the reference’s scale and pushed through the reference’s readout, so a refit matching $\psi(\hat{\phi})$ exactly would reproduce the reference’s predictions exactly. The *warm warp refit* is this same refit started from the reference weights rather than cold. The held-out R^2 column below is the affine R^2 between the refit’s latent and $\psi(\hat{\phi})$ on held-out points, how faithfully the twin reproduces *its own target*, and not predictive accuracy on the assay. It is therefore not comparable with the 0.637 held-out predictive R^2 of the main text, and a value of 1.000000 means the reparameterization was matched, not that the phenotype was predicted perfectly.

The two sets of numbers are not interchangeable. The twin comparison finds more top-3 agreement than independently trained pairs in eleven of twelve cells, but *not* at the best-fitting cell: at width 128 depth 1 the twin’s top-3 set diverges on 46% of sequences against 43% for independently trained pairs. Quoting a twin figure for the main-text population would therefore understate the failure in most cells and overstate it at the one the paper’s headline rests on.

Run 20260830T030657Z_972496a_5da8cb38, git 972496a, config 5da8cb38, 10 seeds, sequence length 9.

H Cold-start recovery, per cell

The defect. The held-out closure experiment (run 20260829T181937Z_b2951ae_9fdd4067, git b2951ae) reported that no cell contains the reparameterized twin, on the strength of a ceiling control that does not do the work it appears to do. Its refit path, `closure_heldout.refit_on_split`, loads the reference weights, then evaluates the loss and snapshots the best iterate *before* entering the

width	depth	held-out R^2	per-instance median ρ	Z to separate
16	1	0.984087 [0.979785, 0.987518]	0.892 [0.877, 0.903]	1 [1, 1]
16	2	0.999241 [0.998708, 0.999619]	0.978 [0.973, 0.983]	43 [25, 60]
16	3	0.999583 [0.999138, 0.999916]	0.984 [0.978, 0.990]	314 [88, 571]
32	1	0.992155 [0.990567, 0.993674]	0.872 [0.847, 0.893]	1 [1, 1]
32	2	0.997328 [0.995451, 0.999059]	0.915 [0.892, 0.937]	87 [14, 193]
32	3	0.999166 [0.998775, 0.999529]	0.923 [0.898, 0.945]	834 [19, 2,136]
64	1	0.998324 [0.997555, 0.998958]	0.899 [0.886, 0.913]	2 [1, 4]
64	2	0.999997 [0.999995, 0.999998]	0.855 [0.832, 0.880]	4,177 [1,320, 7,334]
64	3	0.999996 [0.999993, 0.999998]	0.847 [0.835, 0.860]	7,300 [2,156, 14,437]
128	1	0.999876 [0.999862, 0.999889]	0.923 [0.913, 0.933]	9 [7, 11]
128	2	1.000000 [1.000000, 1.000000]	0.905 [0.888, 0.920]	3,158,530 [1,402,634, 5,004,083]
128	3	1.000000 [1.000000, 1.000000]	0.890 [0.885, 0.895]	87,751,464 [39,140,216, 138,392,841]

Table 9: **A fit and its own twin are far closer than two independently trained fits, which is why the two populations must not be quoted interchangeably.** Z is the number of held-out measurements needed to determine which of the two generated the data at $\alpha = 0.05$ with power 0.8, under the noise model of Appendix F. It is a lower bound: it counts sequencing noise only.

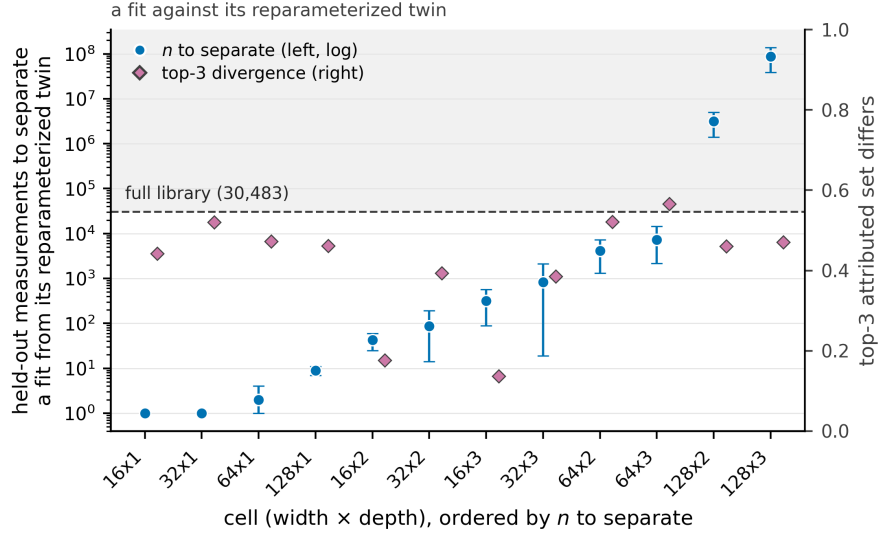


Figure 5: **Separability and top-3 divergence for a fit against its own reparameterized twin, not for independently trained pairs.** The requirement spans nearly eight orders of magnitude here and two cells exceed the full library, where the main-text population separates with at most three measurements in every cell (Figure 3). Left axis logarithmic; the dashed line is the full library.

476 optimisation loop. For the null target, which is the reference’s own latent, the refit therefore begins
477 at the answer. The measured initial loss is 6.7×10^{-19} , already below the 10^{-10} early-stop tolerance,
478 so the loop breaks on its first iteration and returns the reference unchanged.

479 A held-out R^2 of 1.000000 on that target is arithmetic, not evidence. It establishes that Adam does
480 not wander away from a perfect solution. It does not establish that the refit can *find* a solution it does
481 not start at, which is the capability the containment conclusion depends on. Best-iterate selection
482 is what makes the failure silent: because the initial state is already the best iterate ever seen, no
483 subsequent step can displace it, and the procedure reports a perfect score without having searched.

484 **The control that replaced it.** Run 20260829T191427Z_94dd5a2_2267b74e, git 94dd5a2, con-
485 fig 2267b74e, 10 seeds. The null target is refit *without* warm-starting from the reference, from a
486 cold random initialization. The target is known to be in the class, since the reference realises it, so
487 this asks only whether the optimiser can travel to it. Mean initial loss across cells confirms the refit
488 no longer begins at the answer.

width	depth	cold-start null, held-out R^2	warm warp refit, held-out R^2
16	1	0.9988 [0.9982, 0.9993]	0.9748 [0.9669, 0.9809]
16	2	0.9476 [0.9244, 0.9703]	0.9935 [0.9895, 0.9962]
16	3	0.7763 [0.6820, 0.8589]	0.9959 [0.9921, 0.9986]
32	1	0.9994 [0.9990, 0.9997]	0.9749 [0.9708, 0.9789]
32	2	0.8258 [0.7303, 0.9103]	0.9463 [0.9185, 0.9724]
32	3	0.5626 [0.4404, 0.6891]	0.9404 [0.9151, 0.9643]
64	1	0.9971 [0.9955, 0.9984]	0.9804 [0.9696, 0.9893]
64	2	0.7122 [0.5620, 0.8561]	0.9236 [0.8929, 0.9532]
64	3	0.4501 [0.3529, 0.5575]	0.9034 [0.8879, 0.9186]
128	1	0.9969 [0.9962, 0.9975]	0.9885 [0.9868, 0.9900]
128	2	0.8997 [0.8316, 0.9529]	0.9575 [0.9381, 0.9734]
128	3	0.5328 [0.4000, 0.6714]	0.9247 [0.9108, 0.9394]

Table 10: The ceiling that requires search. The cold-start column is an upper bound on what this procedure can say about any target, because its target is in the class by construction. Mean cold-start initial loss ranges from 1.826 to 2.093, against 6.7×10^{-19} for the warm-started control it replaces.

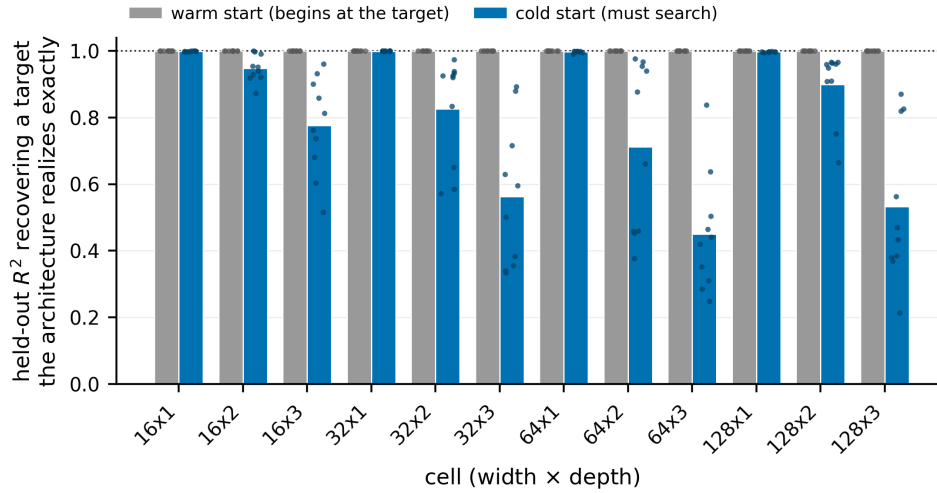


Figure 6: **The warm-start control cannot fail, so it measures nothing.** Held-out R^2 recovering the null target, a function the architecture realizes exactly, with the ten per-seed values overlaid. The warm-started control returns 1.000000 in every cell and every seed, all 120 values being exactly 1.0 rather than approximately, because it is initialized at the reference and its initial loss of 6.7×10^{-19} is already below the 10^{-10} stopping threshold, so the optimization loop never runs and best-iterate selection keeps the initial state. The cold-started arm must actually search for the same target and its per-seed spread shows how much the basin varies: at 128×3 restarts land between 0.21 and 0.87. Warm bars are flat by construction; that flatness is the finding, not a plotting artefact.

489 **What the control shows.** Read by depth: at depth 1 the worst-cell cold-start null reaches 0.9969
490 against 0.9748 for the warm warp refit, so the ceiling sits above the measurement; at depth 2 the
491 worst-cell cold-start null reaches 0.7122 against 0.9236 for the warm warp refit, so the ceiling sits
492 below the measurement; at depth 3 the worst-cell cold-start null reaches 0.4501 against 0.9034 for
493 the warm warp refit, so the ceiling sits below the measurement.

494 At depth 1 the ceiling sits above the measurement, so the comparison is valid there and the warp
495 target is genuinely not reached as well as an in-class one. At depths 2 and 3 the ranking inverts: the
496 warm-started warp refit scores *higher* than a cold-started refit to a target guaranteed to be in the class.
497 That is initialization dominating the result, and no conclusion about what the class contains can be
498 drawn at those depths. The configured pass threshold is 0.999, and 11 of 12 cells fall below it.

499 The honest reading is therefore narrower than the original claim: the refit procedure did not find a
500 member of the class equal to the twin. That is consistent with no such member existing, and equally
501 consistent with one existing that the optimiser failed to reach.